Siqi Dai

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Education

University of Florida, Ph.D. in Machine Learning (expected Jan. 2027) Aug. 2022 –Jan. 2027
Southern University of Science and Technology, Bachelor of Engineering Sep. 2018 –Jun. 2022

Research Interests

Trustworthy multimodal AI: **safety and reliability of multimodal agents**, **privacy risks in vision-language models and conversational agents**, and **adversarial robustness of VLMs**. My work combines risk discovery, rigorous evaluation, and defenses that make deployed AI systems safer.

Publications

- [**CCS 2024**] "GAZEploit: Remote Keystroke Inference Attack by Gaze Estimation from Avatar Views in VR/MR Devices." Hanqiu W. et al. *ACM Conference on Computer and Communications Security (CCS).*
- [**ICC 2025**] "Guided by Noise: Vulnerable Poisoning Attack to Differentially Private Federated Learning." **Siqi Dai** et al. *IEEE International Conference on Communications (ICC).*
- [**ICRA 2024**] "Zero-shot Safety Prediction for Autonomous Robots with Foundation World Models." *IEEE International Conference on Robotics and Automation (ICRA).*
- [**PCIM 2026**] "EMI Diagnosis for Power Converter PCB Layouts based on a Reasoning Aligned Vision Language Model." *PCIM (Power Conversion and Intelligent Motion).*

Research Experience

Graduate Research Assistant, University of Florida – Gainesville, FL Aug. 2022 – Present

- **Privacy in VLMs and agents:** Studied how conversational agents can adaptively elicit sensitive attributes, and how VLMs can make context-aware privacy decisions for image sharing.
- **Robustness of VLMs:** Investigated adversarial jailbreaks of VLMs and defenses that adapt to emerging attacks.
- **Domain-adapted VLM reasoning:** Developed reasoning-aligned VLMs for engineering diagnosis (PCIM 2026).
- **Broader ML security:** Characterized vulnerabilities of differentially private federated learning to targeted poisoning (ICC 2025); contributed to gaze-based keystroke inference research on VR/MR devices (CCS 2024).

Industry Experience

Machine Learning Engineer Intern, ByteDance Inc. E-commerce Group – San Jose, CA May 2026 – Aug. 2026

- Designed and deployed a **self-improving agent system** that diagnoses and refines its own risk-detection workflows from **execution traces**, using MCP-based tool interfaces; adopted across **100+ workflows and 40+ users**.
- Trace-driven refinement of prompts and context, code, and data-processing logic raised risk-detection recall from **63.1% to 94.3%** while reducing the **false-positive rate by 87%**.
- Constrained self-modification with schema/tool checks and protected-node constraints, and used **LLM-as-a-Judge** to accept candidate changes based on groundedness and reasoning quality.
- Built automated fault recovery that distinguishes unavailable data from valid results, reducing execution failures from **80% to near zero** on targeted high-failure workflows.
- Integrated the agent into an **award-winning human-in-the-loop (HITL) initiative**, which ranked **5th among 40 projects** in a department-wide competition within ByteDance's Governance and Experience (GNE) organization.

Selected Research Projects

ShareGuard: Context-Aware Multimodal Privacy Review

- Addressed **recipient-induced decision bias** in VLM-based privacy judgments by decomposing review into sensitive-content extraction, audience-boundary inference, and recipient matching.
- Across **five VLMs on 160 cases**, outperformed text-only, image-only, and rule-lookup baselines; improved **GPT-4.1 sharing-decision accuracy from 75.0% to 91.2%**.

PrivacyProbe: Automated Privacy Red Teaming for Conversational AI

- Studied how conversational agents can adaptively elicit sensitive attributes; proposed an attacker that uses **structured conversational memory** to track inference evidence and guide questioning across four service personas.
- On **30 synthetic profiles, 19 attributes, and 15-turn conversations**, achieved **75.3% macro-averaged inference accuracy** in the life-coach setting, exceeding non-adversarial and passive baselines by **29.2 and 49.5 percentage points**; ablations isolated the contributions of memory and target selection.

Adversarial Detection and Jailbreak Defense for VLMs

- Proposed a **quantization-based defense** against adversarial VLM jailbreaks, achieving a **95% defense success rate**, and a **ViT–ResNet meta-learning detector** that adapts to emerging adversarial inputs.
- Improved robustness via **cross-modal gradient alignment** between visual and text embeddings; evaluated transferability across multiple open-source VLMs.

Reasoning-Aligned VLM for PCB EMI Diagnosis *PCIM 2026*

- Adapted **Qwen2.5-VL** to electromagnetic interference (EMI) analysis via **multimodal supervised fine-tuning** with LoRA, on a curated **multi-image, multi-turn instruction dataset** spanning **38 PCB designs and 254 images**.
- Designed a **three-stage VLM reasoning pipeline** that grounds predictions in retrieved layout guidance and device metadata.

Targeted Poisoning in Federated Learning *ICC 2025*

- Developed a stealthy targeted model-poisoning attack that reduces target-class accuracy to **35%** while limiting non-target degradation, using **KL divergence** over weight distributions to evade **Krum** and k**-out-of-**n defenses.

Gaze-Based Keystroke Inference on Apple Vision Pro *CCS 2024, co-author*

- Implemented an ML pipeline using **3D gaze estimation, typing-session detection, and top-**K **prediction** to infer keystrokes from avatar gaze, contributing to an **Apple-confirmed vulnerability**.

Technical Skills

Research Areas: Multimodal Agents, VLMs, Privacy, Adversarial Robustness, Federated Learning, LLM Evaluation
Methods & Tools: Python, PyTorch, Hugging Face Transformers, PEFT/LoRA, Multimodal SFT, Meta-Learning, RAG, MCP, AWS, Linux, Git